

Spotlight: Systematic Multi-Asset Funds with News-Sentiment Analytics

Nuo Chen (z5640476)

FINS5545 Financial Market Data Literacy - Part B

Abstract

Spotlight offers twelve systematic funds built from 50 US equities and 10 cryptocurrencies. Over a three-year out-of-sample backtest the best fund, Combined Maximum-Sharpe, earns a Sharpe ratio of 0.98, but the more durable result is that equal weighting beats every optimised equity fund (0.82 against 0.72 for the closest optimised rival). The methods that lean hardest on estimated inputs are the most concentrated, the most heavily traded, and the worst out of sample. A sector news-sentiment index built from 146,830 headlines tracks tone convincingly but does not predict returns: sentiment correlates with same-day sector returns in seven of ten sectors and with next-day returns in one. A sentiment tilt therefore reduces the Sharpe ratio of both equity funds; making its direction an estimate rather than an assumption recovers the loss, though bootstrap intervals for that improvement span zero. The funds, fact sheets and analytics ship as a deployed Streamlit application.

Contents

Abstract.....	1
Contents.....	1
1. The funds and the backtest design.....	2
How often the funds trade	3
2. Out-of-sample results and fund fact sheets	4
3. The news-sentiment index	7
4. Fusion: does sentiment improve the funds?	9
Is the signal there at a slower frequency?	11
5. Extensions.....	12
5.1 A finance lexicon for VADER	12
5.2 A tilt that learns its own direction.....	13
5.3 Was the winner found or fitted?	15
5.4 Fear and greed as a timing overlay	16
6. The app and the investor journey	17

7. Critical reflection and recommendations	17
References	18
Appendix	19
Appendix A - The four objectives.....	19
Appendix B - Supporting exhibits	20
Appendix C - The figure design system.....	21
Appendix D - Rolling against expanding estimation windows	21
Appendix E - Does correcting the covariance fix the result?	22

Right-click and choose Update Field in Word to populate this.

1. The funds and the backtest design

Spotlight offers 12 funds. Each is one asset family - the 50 US equities, the 10 cryptocurrencies, or the two combined - run through one optimisation method, because a family-method pair is what an investor buys and what a fact sheet covers. The four methods are equal weight, minimum variance, maximum Sharpe (mean-variance tangency, following Markowitz 1952), and risk parity (Maillard, Roncalli and Teiletche 2010). All are long only and fully invested, so weights are non-negative and sum to one.

Weights come from a walk-forward backtest. At each rebalance the optimiser sees a rolling window of the previous 252 trading days and nothing after it, and the resulting weights are held until the next rebalance, drifting with prices in between as a real fund does. Rebalancing is monthly, on the first trading day of each month, giving 36 rebalances. The out-of-sample record runs from 2021-01-01 to 2023-12-31 (1095 trading days for the equity and combined funds); it begins after the first full estimation window, not on the first date in the data.

Two conventions are stated rather than assumed. The risk-free rate is zero, so the Sharpe ratio (Sharpe 1966, 1994) is the annualised mean return divided by annualised volatility. Headline results are gross of transaction costs; a 10 basis point charge on one-way turnover is reported alongside as a robustness layer. Equity and combined funds annualise on 252 days and crypto-only funds on 365, matching each calendar.

The four objectives are stated formally in Appendix A, with every symbol defined.

Three design choices deserve defending. The estimation window rolls rather than expands, so the covariance matrix reflects the most recent year of trading rather than an average of the whole history, which trades a noisier estimate for responsiveness to a change in regime. Rebalancing monthly balances the same tension the other way: often enough to act on a changed covariance estimate, rarely enough not to pay for chasing noise. Long-only weights match what the product is and bound the damage an extreme estimate can do, since no weight can go below zero however attractive a short looks. Jagannathan and Ma (2003) show that a no-

short-sale constraint acts on a minimum-variance problem in the same way as shrinking the most extreme covariance estimates, so the constraint buys statistical stability as well as investability. Appendix D tests that trade-off: an expanding window lifts the equity funds (+0.110 on minimum variance) and costs the flagship Combined Maximum-Sharpe fund 0.583, because it never forgets the crypto regime that ended in 2022.

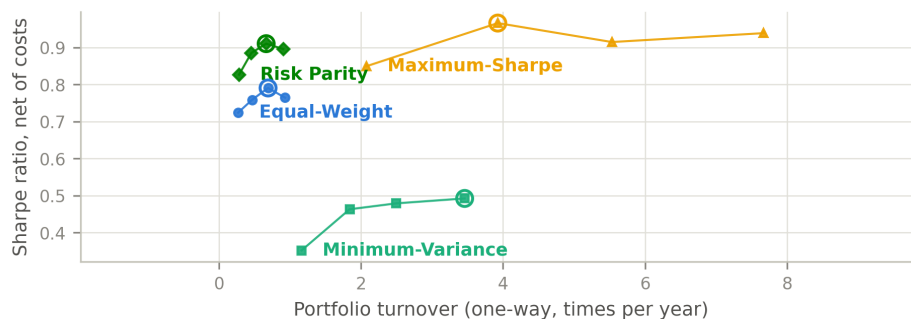
The absence of look-ahead is asserted rather than argued. `tests/test_no_lookahead.py` runs each fund on the full sample and again on a sample truncated at 30 June 2022 and requires the overlapping returns and weights to match exactly; a backtest that used future data could not pass.

How often the funds trade

Monthly rebalancing is a choice, so the alternatives were measured. Weekly, fortnightly, monthly and quarterly schedules were run for each method on the combined universe, gross and net of costs.

How often should the fund trade?

Out-of-sample Sharpe ratio of the combined funds under four rebalance schedules, net of 10 bp of one-way turnover, against the turnover each schedule generates. Points run weekly, fortnightly, monthly, quarterly; the ring marks each method's best schedule. Quarterly is worst everywhere; beyond that the best schedule depends on how much the optimiser moves the weights.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 1: Sharpe ratio net of 10 bp of one-way turnover against annual turnover, for four rebalance schedules across the four combined funds. Out-of-sample, 2021-2023.

Best net-of-cost schedule by method: Equal-Weight fortnightly; Maximum-Sharpe monthly; Minimum-Variance weekly; Risk Parity fortnightly.

Quarterly rebalancing is worst for every method, which says the cost of holding a stale covariance estimate for three months exceeds the trading it saves. Above that floor the schedules separate by how much each method's weights move. Equal weight and risk parity barely trade whatever the schedule, so their fortnightly advantage is small. Maximum Sharpe turns over several times a year, so it is the fund where the choice bites, and monthly is its best net-of-cost schedule: weekly gives it a fresher estimate but the extra turnover is not repaid. Monthly is kept because it is the best net outcome for the fund most sensitive to the decision and within a few hundredths of the best for the rest.

2. Out-of-sample results and fund fact sheets

Table 1: Out-of-sample performance of all twelve funds. Annualised return is geometric; the Sharpe ratio uses the arithmetic mean with a zero risk-free rate. Equity and combined funds annualise on 252 days, crypto-only on 365. Sample 2021-2023.

fund	ann return	ann vol	sharpe	max drawdown	growth of 1
Combined Maximum-Sharpe	0.240	0.252	0.981	-0.258	1.902
Combined Risk Parity	0.140	0.162	0.888	-0.195	1.478
Combined Equal-Weight	0.151	0.216	0.761	-0.279	1.524
Combined Minimum-Variance	0.054	0.128	0.477	-0.154	1.171
Crypto Risk Parity	0.443	0.799	0.862	-0.799	3.004
Crypto Equal-Weight	0.405	0.819	0.829	-0.816	2.774
Crypto Minimum-Variance	0.320	0.650	0.753	-0.758	2.301
Crypto Maximum-Sharpe	-0.235	0.798	0.065	-0.889	0.447
Equity Equal-Weight	0.126	0.161	0.819	-0.202	1.427
Equity Risk Parity	0.099	0.145	0.723	-0.185	1.326
Equity Maximum-Sharpe	0.092	0.182	0.573	-0.256	1.299
Equity Minimum-Variance	0.054	0.127	0.479	-0.153	1.171

Equal weighting earns the highest Sharpe ratio of the four equity funds at 0.819, ahead of risk parity at 0.723, maximum Sharpe at 0.573 and minimum variance at 0.479. The ordering is not one story but two, and telling them apart matters more than the ranking does.

The first is sample noise. Every rebalance asks 252 days of returns for 50 expected returns and a 50-by-50 covariance matrix, a thin sample for that many quantities. Michaud (1989) calls the consequence error maximisation: an optimiser given noisy inputs overweights precisely those assets whose expected returns the sample has overstated and whose risks it has understated, so estimation error is amplified rather than diversified away. Maximum Sharpe holds 6 of the 50 stocks and rewrites 34% of its portfolio at the average rebalance; minimum variance holds 18 and rewrites 15%. Equal weighting estimates nothing and never trades, and risk parity, which needs variances but no expected returns, changes 2% of its weights a month. Chopra and Ziemba (1993) find errors in expected returns roughly an order of magnitude more costly than errors in variances, which is consistent with maximum Sharpe - the one method here needing a mean estimate - being both the most concentrated fund and the most heavily traded. That turnover and concentration are not the optimisation working but a symptom of how noisy its inputs are.

How small is quantified by DeMiguel, Garlappi and Uppal (2009), who evaluate fourteen mean-variance models and their estimation-error corrections against 1/N across seven datasets and find none of them consistently better on Sharpe ratio, certainty-equivalent return or turnover. Calibrating to US equities, they estimate that a sample-based mean-variance strategy needs an estimation window of roughly 6,000 months to beat 1/N on a 50-asset portfolio. This fund holds 50 assets and estimates on 12 months. On their arithmetic the equity results here are not a

surprising outcome but the expected one, and the question is less why optimisation lost than why anyone would have expected it to win at this sample size.

The second force is not error at all. Minimum variance ranks last on the Sharpe ratio while achieving exactly what it was built to achieve: its 12.7% annualised volatility is the lowest of the four funds and its 15.3% maximum drawdown the shallowest. The objective never mentions return, so reading its last place as a failure of optimisation would be wrong. The fund solved its problem; the Sharpe ratio scores a different one.

That is a statement about who the fund is for rather than about whether it works. An investor ranking funds on risk-adjusted return should not hold it - at +5.4% a year it is the weakest equity fund on offer. An investor who will sell in a drawdown should, because it is the fund that fell least, and the return an investor keeps is the return they hold through. Spotlight therefore lists all twelve funds rather than only the top of the Sharpe table, and the allocation tool exists so that a user can express which of those two investors they are. A single ranking cannot carry that information, which is why the fact sheet shows the drawdown next to the growth line.

Whether that first force really is sample noise is testable, and Section 5.4 tests it by correcting the covariance estimate and re-running every fund.

One number in the table needs reading carefully. Crypto Maximum-Sharpe reports a positive Sharpe ratio of 0.065 alongside an annualised return of -23.5%. Both are correct. The Sharpe ratio uses the arithmetic mean, +5.2% a year, while the headline return is geometric - what a dollar actually compounds to. The two diverge by roughly half the variance. At 79.8% annualised volatility that is 0.798 squared over two, or 31.9%, against an observed gap of 28.8%; the approximation overshoots by about 3.1% because it drops the higher-order terms that matter at this volatility. The direction is what counts: a fund can average a gain every day and still destroy capital.

Growth of \$1: combined equity and crypto funds

Value of \$1 invested at the first live backtest date, 04 Jan 2021 to 29 Dec 2023. Weights are re-optimised on the first trading day of each month from the prior year of returns only.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

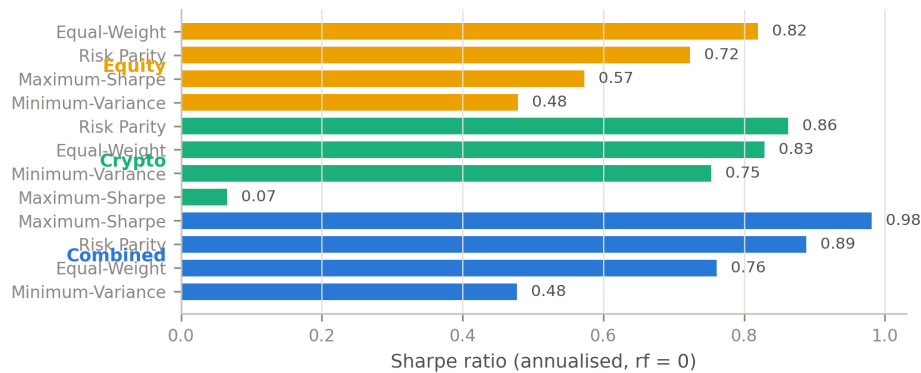
Figure 2: Growth of \$1 invested at the first live backtest date in the four combined equity-plus-crypto funds.

The lines separate in the first quarter of 2021 and never reconverge. Maximum Sharpe pulls away early because its concentration paid during the crypto rally, keeps most of that lead

through the 2022 drawdown, and ends well ahead. The path matters as much as the endpoint: the fund that ends highest also fell furthest, so choosing between it and risk parity is a choice about tolerance rather than about which manager is better.

Sharpe ratio across funds and methods

Annualised Sharpe ratio (risk-free rate assumed zero) over each fund's out-of-sample period. Equity and combined funds annualise on 252 days, crypto-only funds on 365.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 3: Sharpe ratio across all twelve funds, grouped by asset family.

The crypto funds top the risk-adjusted table despite their drawdowns, because 2021 to 2023 was a period when crypto returns were large enough to compensate for their volatility - a statement about this sample, not the asset class. Within the equity funds the ordering runs against what optimisation theory promises: equal weighting at 0.819 beats every optimised rule. The combined funds sit between the two families, which is what diversification should produce, though it also means mixing crypto with equities lowered the Sharpe ratio against holding crypto alone over this window.

Drawdown: Combined Maximum-Sharpe

Percentage below the running peak of growth of \$1, 04 Jan 2021 to 29 Dec 2023. Deepest drawdown -25.8% on 30 Sep 2022.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 4: Drawdown of the Combined Maximum-Sharpe fund from its running peak.

The fund lost 25.8% from its peak and took most of a year to recover. An investor who bought at the high watched a quarter of their capital disappear before any of the eventual return arrived, which is what a Sharpe ratio of 0.98 conceals. The app's fact sheet therefore shows drawdown alongside the growth line: together they describe the product, and either alone misleads.

Portfolio weights over time: Combined funds

Target weights at each monthly rebalance, by sector, 04 Jan 2021 to 01 Dec 2023. Same universe and dates under each objective; crypto is the dark band.

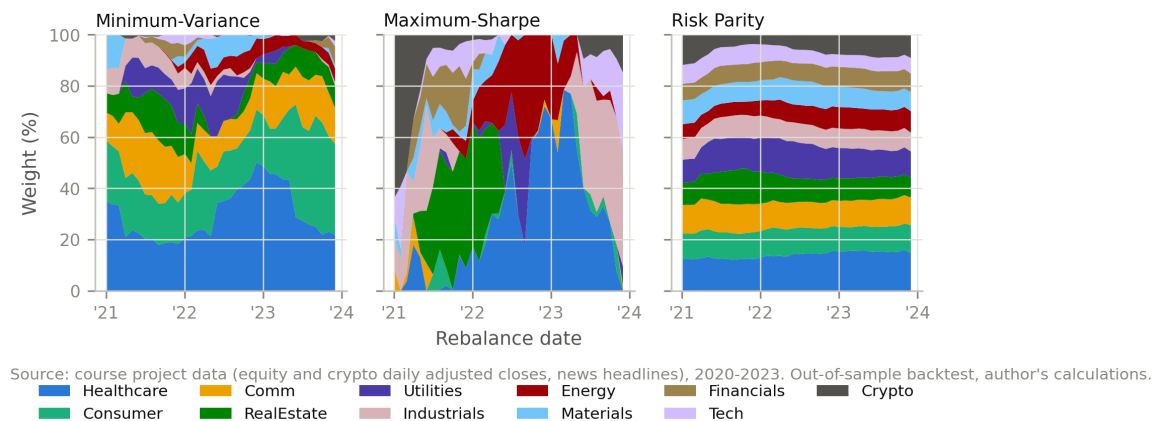


Figure 5: Target weights at each monthly rebalance for three combined funds, aggregated to sector and stacked to 100%. Crypto is the dark band.

The three panels show one universe and one set of dates producing three different funds. Minimum variance concentrates in healthcare and consumer staples; maximum Sharpe swings between sectors, chasing mean estimates that change month to month. Risk parity is nearly static, because relative volatilities are far more stable than means. Average crypto weight quantifies that: 16.7% under equal weighting by construction, 9.4% under maximum Sharpe, 7.0% under risk parity, 0.6% under minimum variance. Equalising risk contributions prices a volatile asset rather than excluding it: crypto keeps a share proportionate to its risk, where minimising variance drops it outright. The stability of each panel pictures how much estimation error each objective admits.

3. The news-sentiment index

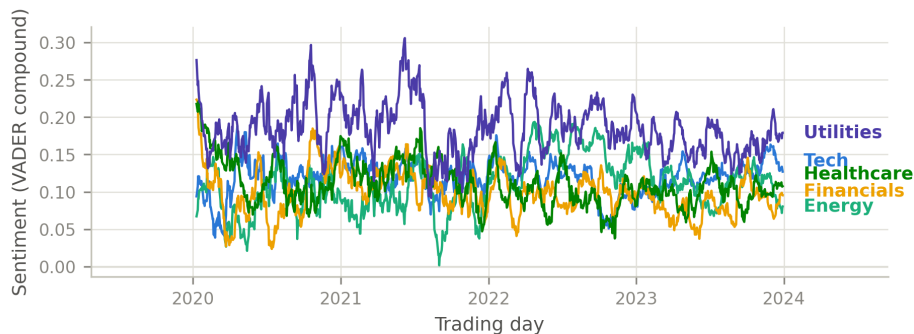
Every one of the 146,830 aligned headlines is scored with VADER (Hutto and Gilbert 2014). The text is scored raw: VADER reads capitalisation, punctuation, degree modifiers and negation, so lower-casing or stripping punctuation would remove the evidence the model is built on. Scores are averaged within a ticker-day, then equally weighted across the five tickers in each sector, so that one heavily covered name cannot speak for its sector.

Ticker-days with no headline are dropped from the sector average rather than scored as zero. A zero would pull the index toward neutral on quiet days and would confuse an absence of news with balanced news, and Part A found multi-month collection blackouts for two tickers that a zero-fill would have turned into fake neutral sentiment. The daily coverage count is published beside the index so thin days are visible.

The index is lagged one trading day after alignment. A decision on day t therefore uses only sentiment from $t-1$ or earlier: a Saturday or Monday headline, both aligned to Monday, is first usable for Tuesday's trade.

News-sentiment index by equity sector

Equal-weight mean VADER compound score across the five tickers in each sector, 21-day rolling mean, 02 Jan 2020 to 29 Dec 2023. Scores run from -1 (most negative) to +1; ticker-days with no headlines are excluded. Every sector averages positive - headline sentiment carries a well-known optimistic bias.



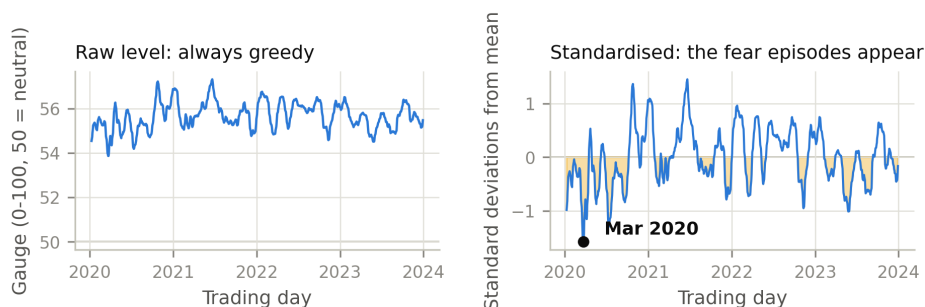
Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 6: News-sentiment index for five equity sectors: equal-weight mean VADER compound score, 21-day rolling mean, 2020-2023.

Every sector's index sits above zero for most of the sample. That is a property of headline writing rather than of the market: coverage is written to attract readers and leans positive, so only the index's movement carries information. Utilities runs highest at +0.179 and financials lowest at +0.090, a difference that reflects the vocabulary of each sector's news - dividends and stability against litigation and rates - more than any difference in performance. Energy is the one sector whose index moves visibly with its fundamentals, rising through 2022 as the sector's returns did.

Fear and greed across the whole equity market

Sentiment averaged over all 50 stocks. On the left the raw 0-100 gauge, above neutral on 99% of days because headline sentiment has a positive baseline. On the right the same series standardised, which is what makes the fear episodes visible. Both panels show a 21-day rolling mean.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 7: Market-wide fear and greed gauge: sentiment averaged across all 50 equities, shown as the raw 0-100 level and standardised.

The raw gauge sits above neutral on 99% of days, so read as a level it would have called the market greedy throughout the COVID crash. Standardising removes that positive baseline and the episodes appear immediately. The deepest trough is mid-March 2020, the week of the crash itself, and the second is early December 2021, when the Omicron variant was announced. That the two largest negative excursions in a purely textual series fall on the two events a reader

would name unprompted is the strongest available evidence that the index measures something real; the search-based index of Da, Engelberg and Gao (2015) behaves the same way.

Table 2: Coverage and neutral share of the sentiment model by sector.

sector	n headlines	n tickers	mean compound	neutral share	n days with news
Comm	11783	5	0.099	0.523	997
Consumer	25877	5	0.094	0.517	998
Energy	19365	5	0.106	0.438	996
Financials	18269	5	0.090	0.493	998
Healthcare	14697	5	0.107	0.501	997
Industrials	13156	5	0.090	0.498	998
Materials	5393	5	0.096	0.576	940
RealEstate	5339	5	0.133	0.533	940
Tech	26632	5	0.116	0.503	1006
Utilities	6319	5	0.179	0.395	962

Between 40% and 58% of headlines score neutral depending on the sector. A neutral score is not a finding of balance; it means the model found no word it recognised. VADER's lexicon was built for social media, and Loughran and McDonald (2011) show that dictionaries assembled outside a financial context misclassify financial language systematically; Section 5.1 measures how much of this band is that gap rather than genuine balance. Headlines are also weaker than the articles beneath them, and work extracting firm-level information from news generally scores the full story rather than the headline (Tetlock, Saar-Tsechansky and Macskassy 2008). The index reads as the tone of coverage, and any trading use of it must survive half its inputs being silent.

4. Fusion: does sentiment improve the funds?

The baseline fusion is a cross-sectional sector tilt. At each rebalance the lagged sector index becomes a z-score across the ten sectors, and each equity weight is multiplied by $(1 + \lambda z)$ for its sector, then clipped at zero and renormalised so the fund stays long-only and fully invested. The tilt never touches the covariance estimate, so any difference comes from the tilt alone. Sentiment applies to equities only, because the cryptocurrencies carry no news.

Table 3: Equity funds before and after the sentiment tilt, out of sample.

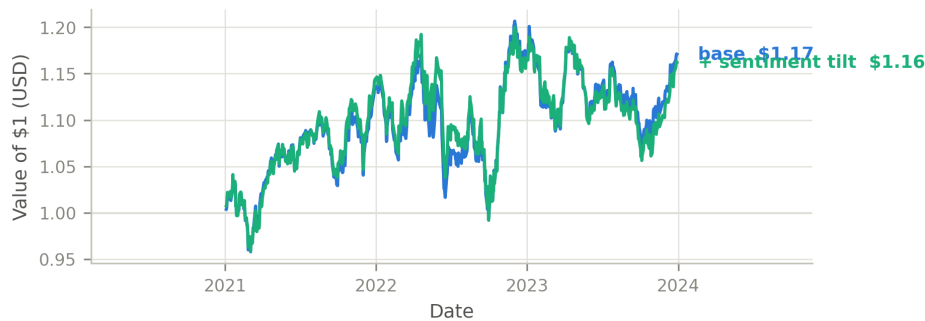
fund	ann return	ann vol	sharpe	max drawdown	growth of 1	avg turnover
Equity Minimum-Variance (base)	0.054	0.127	0.479	-0.153	1.171	0.151
Equity Minimum-Variance + sentiment	0.052	0.128	0.457	-0.168	1.162	0.215
difference	-0.003	0.001	-0.022	-0.016	-0.009	0.064
Equity Maximum-Sharpe (base)	0.092	0.182	0.573	-0.256	1.299	0.331
Equity Maximum-Sharpe + sentiment	0.081	0.186	0.509	-0.270	1.260	0.365
difference	-0.011	0.004	-0.064	-0.013	-0.039	0.034

The tilt makes both funds worse. Minimum variance falls from 0.479 to 0.457 and maximum Sharpe from 0.573 to 0.509, while turnover rises in both. That combination - lower return, higher trading - is what a signal with no predictive content looks like when it is used to move weights:

the tilt adds noise to a portfolio that was already as good as its inputs allowed, and charges for the privilege.

Sentiment fusion: Equity Minimum-Variance

Growth of \$1 for the base fund and the same fund with the sector sentiment tilt applied at each rebalance. The tilt reads sentiment lagged one trading day, so no headline is used before it is public.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 8: Growth of \$1 for the Equity Minimum-Variance fund with and without the sentiment tilt.

Sentiment fusion: Equity Maximum-Sharpe

Growth of \$1 for the base fund and the same fund with the sector sentiment tilt applied at each rebalance. The tilt reads sentiment lagged one trading day, so no headline is used before it is public.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 9: Growth of \$1 for the Equity Maximum-Sharpe fund with and without the sentiment tilt.

The diagnostics explain the direction of the result. The correlation between the lagged sector index and the sector return it can be traded on is negative in 8 of the ten sectors, with a mean of -0.0205.

Table 4: Correlation between the lagged sector sentiment index and the equal-weight sector return on the day the signal becomes usable.

sector	n days	corr lagged sentiment vs return	mean return high sentiment	mean return low sentiment	spread bps per day
Consumer	1002	0.028	0.001	-0.000	10.118
Materials	1002	0.020	0.002	-0.000	17.434
Tech	1005	-0.009	0.000	0.002	-19.545
Healthcare	1002	-0.012	0.000	0.001	-9.295
Comm	1002	-0.014	0.000	0.000	-0.748
Financials	1002	-0.021	0.001	0.001	0.048

RealEstate	1002	-0.037	-0.000	0.001	-14.500
Industrials	1002	-0.039	-0.001	0.002	-27.196
Utilities	1002	-0.052	-0.001	0.001	-16.246
Energy	1002	-0.071	-0.001	0.003	-31.842

A negative correlation is consistent with three different stories - a reversal, news already in the price, or noise - so sentiment on day t was correlated against the sector return on the day before, the same day, and the day after, with a p-value for each.

Table 5: Sector sentiment on day t against the sector return on t-1, t and t+1, with Pearson p-values. Raw daily index, 2020-2023.

sector	corr prev day	p prev day	corr same day	p same day	corr next day	p next day
Tech	0.032	0.316	0.121	0.000	-0.031	0.326
Energy	0.034	0.286	0.120	0.000	-0.027	0.393
Industrials	0.113	0.000	0.119	0.000	-0.020	0.523
Financials	0.076	0.017	0.118	0.000	0.048	0.128
Utilities	0.053	0.098	0.097	0.003	-0.088	0.006
Materials	0.061	0.061	0.096	0.003	0.015	0.656
Consumer	0.015	0.632	0.082	0.010	0.001	0.972
RealEstate	0.040	0.223	0.061	0.063	0.002	0.949
Healthcare	-0.002	0.958	0.053	0.097	-0.023	0.466
Comm	-0.014	0.656	0.018	0.578	-0.003	0.924

Mean correlations: +0.0885 same day (7 of 10 significant at 5%), -0.0127 next day (1 of 10), and +0.0408 the day before (2 of 10).

The same-day correlation is the large one. Sentiment moves with the sector return on the day it is published, +0.0885 on average and significant in 7 of ten sectors. The next-day correlation is -0.0127 and significant in 1. The news is real information and it is related to prices, but by the time a headline has been published and a trading day has closed, that information is in the price. This is the 'already priced' reading rather than a reversal: a genuine reversal would show up as a large, significant NEGATIVE next-day coefficient, and it does not. The negative signs reported in the diagnostics above are small and mostly indistinguishable from zero.

That conclusion is the one an efficient market would predict at this horizon (Fama 1970), and it explains the fusion result without needing the tilt to be badly built. Tetlock (2007) finds that the price pressure associated with media pessimism reverts within a few trading days, so even where news tone does move prices the effect is short-lived relative to a monthly rebalance. There is nothing left in daily headline sentiment to trade.

Is the signal there at a slower frequency?

Absorption by the daily close does not rule out a slower effect, so sentiment averaged over the past k trading days was tested against the return over the next k days, for k of 1, 5, 21 and 63. Observations are sampled every k days rather than every day, because overlapping windows share data and would inflate significance badly at the longer horizons, which is the mechanism behind the spurious long-horizon predictability documented by Boudoukh, Richardson and Whitelaw (2008). Each horizon is run twice: on raw sector returns, and cross-sectionally with the day's mean across sectors removed from both sides, which is what the tilt actually bets on.

Table 6: Past-k-day sector sentiment against the next-k-day sector return, non-overlapping observations. The corrected column applies a Bonferroni threshold across the eight tests reported here.

horizon days	variant	n obs	correlation	p value	significant 5pct	significant corrected
1	absolute	9821	-0.013	0.199	False	False
1	cross-	9821	-0.022	0.029	True	False

	sectional					
5	absolute	1990	-0.016	0.470	False	False
5	cross-sectional	1990	-0.023	0.311	False	False
21	absolute	460	-0.088	0.060	False	False
21	cross-sectional	460	-0.069	0.141	False	False
63	absolute	140	-0.149	0.080	False	False
63	cross-sectional	140	-0.121	0.153	False	False

The correlation is negative at every horizon and its magnitude grows with it, from -0.0130 at one day to -0.1487 at 63 days, the pattern a slow effect would leave. Nothing else supports it. Independent observations collapse from 9,821 to 140 as the horizon lengthens, so the confidence around each estimate widens accordingly: 1 of 8 tests clears 5% uncorrected and none survives the correction for having run eight. One nominal p-value of 0.029 out of eight tests is what chance delivers. Harvey, Liu and Zhu (2016) argue that conventional thresholds are far too lenient once the number of specifications tried is counted, and the Bonferroni threshold applied here is the most conservative version of that argument.

The honest reading is that this is a null result with a suggestive shape, and that separating a weak slow effect from noise would need a longer sample than three years rather than a cleverer test on this one. The search was conducted over eight specifications and that is reported here rather than only the specification that came closest.

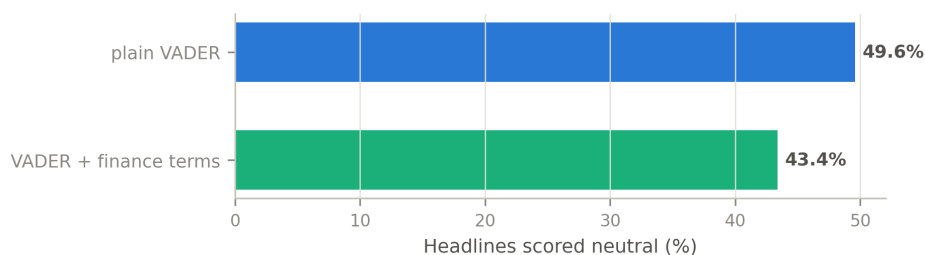
5. Extensions

5.1 A finance lexicon for VADER

Only 1,831 of the 33,033 distinct tokens in the headline corpus carry a VADER score, and 49.6% of headlines score neutral. The gaps are not obscure: 'earnings' appears 11,870 times and 'beat' 2,083, and VADER scores neither. This is the mismatch Loughran and McDonald (2011) document for general-purpose dictionaries applied to financial text, and their remedy - a lexicon derived from the corpus actually being scored - is the one followed here. 46 finance terms were added, selected by frequency from the headlines themselves rather than from memory, each with a written rationale. Terms whose sign reverses with context - 'cut', 'hike', 'buy' - were deliberately excluded and are listed with their reasons.

The finance lexicon cuts VADER's false neutrals

Share of the 146,830 aligned headlines scoring within +/-0.05 - VADER's own neutral band - before and after merging 46 finance terms into the lexicon. Only 1,831 of the 33,033 distinct headline tokens carry a VADER score at all.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 10: Share of headlines scoring inside VADER's neutral band, before and after adding the finance terms.

The neutral share falls to 43.4% and 16.1% of headlines are rescored.

The headlines being missed are the most informative ones. 'AbbVie Q4 EPS \$2.21 Beats \$2.19 Estimate' scores exactly zero under plain VADER, because neither 'beats' nor 'estimate' is in a lexicon built for social media. Adding 46 market terms cuts the neutral share from 49.6% to 43.4% and rescores 16.1% of headlines, and the dispersion of scores rises from 0.288 to 0.317.

A lower neutral share is not automatically a better signal. The extension makes the index a more faithful description of tone, which is what it was built for, but it also fires on headlines whose sentiment is not about the company: 'Inovio's Surge on a Coronavirus Vaccine Was Just Speculation' scores positive because 'surge' is now scored, though the sentence is sceptical. Whether the added accuracy helps depends entirely on what the index is used for, and the next section shows that it helps one use and hurts another.

5.2 A tilt that learns its own direction

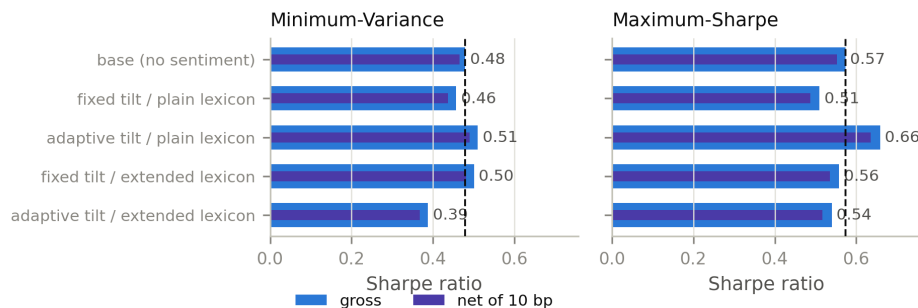
Because the naive tilt leans the wrong way, the direction is made an estimate rather than an assumption. At each rebalance the pooled correlation between lagged sentiment and sector returns is re-estimated on the previous 252 trading days only, and the tilt leans whichever way that window indicates. Hard-coding the reverse would have fitted the answer to data already seen; this does not, and the truncation test covers it.

Table 7: Every fusion variant against its base fund, gross and net of 10 bp of one-way turnover.

fund	variant	sharpe	sharpe net costs	avg turnover
Equity Minimum-Variance	base (no sentiment)	0.479	0.465	0.151
Equity Minimum-Variance	fixed tilt / plain lexicon	0.457	0.437	0.215
Equity Minimum-Variance	adaptive tilt / plain lexicon	0.510	0.490	0.218
Equity Minimum-Variance	fixed tilt / extended lexicon	0.501	0.482	0.212
Equity Minimum-Variance	adaptive tilt / extended lexicon	0.387	0.367	0.220
Equity Maximum-Sharpe	base (no sentiment)	0.573	0.552	0.331
Equity Maximum-Sharpe	fixed tilt / plain lexicon	0.509	0.487	0.365
Equity Maximum-Sharpe	adaptive tilt / plain lexicon	0.658	0.636	0.354
Equity Maximum-Sharpe	fixed tilt / extended lexicon	0.558	0.535	0.366
Equity Maximum-Sharpe	adaptive tilt / extended lexicon	0.540	0.517	0.363

Sentiment variants against the base fund

Out-of-sample Sharpe ratio of each equity fund under the fixed tilt and the adaptive-direction tilt, on the plain and the finance-extended lexicon. Solid bars are gross; the darker inset charges 10 bp of one-way turnover at each rebalance. The dashed line marks the base fund.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 11: Sharpe ratio of each fusion variant against the base fund, gross and net of transaction costs.

A higher point estimate is not the same as an improvement. The difference in Sharpe ratios between each tilt and its base fund was resampled with a stationary bootstrap (Politis and Romano 1994) of 5,000 draws and a mean block length of 21 trading days, which keeps the two series paired and preserves their autocorrelation. Ledoit and Wolf (2008) show that a paired bootstrap of this form is the appropriate test for a difference in Sharpe ratios when returns are autocorrelated and non-normal, as daily fund returns are.

Table 8: Bootstrap confidence intervals for the change in Sharpe ratio relative to the base fund.

fund	variant	difference	ci low	ci high	p value	significant 5pct
Equity Minimum-Variance	adaptive tilt / plain lexicon	0.031	-0.133	0.204	0.749	False
Equity Minimum-Variance	fixed tilt / plain lexicon	-0.022	-0.183	0.138	0.800	False
Equity Maximum-Sharpe	adaptive tilt / plain lexicon	0.085	-0.081	0.254	0.311	False
Equity Maximum-Sharpe	fixed tilt / plain lexicon	-0.064	-0.208	0.087	0.408	False

The adaptive tilt raises the point estimate on both funds, by +0.085 of a Sharpe point at best. Every bootstrap interval spans zero. On three years of daily data the noise around a Sharpe difference is wider than the difference itself, so the improvement is not established, however it looks in the table. Reporting it as a result would be claiming more than the sample supports.

What can be said is narrower: making the direction an estimate rather than an assumption removes the damage the fixed tilt does, without look-ahead. Establishing that it adds value would need a longer sample or a second market.

The two extensions do not stack. The finance lexicon helps the fixed tilt and hurts the adaptive one. The diagnostics explain why: the extended lexicon weakens the very correlation the adaptive tilt trades against, moving the mean from -0.0205 to -0.0156 and the count of negative

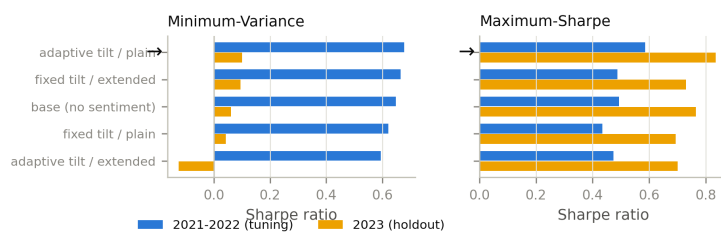
sectors from eight to seven. Part of what the adaptive tilt was exploiting was VADER's blind spots rather than a property of the news, which is a reason to treat its advantage cautiously even before the confidence intervals are considered.

5.3 Was the winner found or fitted?

Comparing five variants over one period and keeping the best is itself a choice fitted to that period, and the resulting performance is biased upward by the search rather than by any property of the winner (White 2000). The variants were therefore re-ranked on a 2021-2022 discovery window alone, and the selected variant's 2023 performance reported, 2023 having played no part in the choice.

What tuning picks, and whether it survives

Each variant's Sharpe ratio over the 2021-2022 discovery window and over 2023, which was never used to choose between them. The variant tuning would have selected is marked. A variant that wins on the left and collapses on the right was fitted, not found.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations. Arrow marks the variant tuning selects.

Figure 12: Sharpe ratio in the 2021-2022 tuning window against the untouched 2023 holdout, for each fusion variant.

Table 9: Discovery-window ranking and holdout outcome for each variant.

fund	variant	discovery sharpe	holdout sharpe	selected by tuning
Equity Minimum-Variance	adaptive tilt / plain lexicon	0.677	0.100	True
Equity Minimum-Variance	fixed tilt / extended lexicon	0.664	0.094	False
Equity Minimum-Variance	base (no sentiment)	0.648	0.060	False
Equity Minimum-Variance	fixed tilt / plain lexicon	0.622	0.042	False
Equity Minimum-Variance	adaptive tilt / extended lexicon	0.595	-0.126	False
Equity Maximum-Sharpe	adaptive tilt / plain lexicon	0.585	0.835	True
Equity Maximum-Sharpe	base (no sentiment)	0.494	0.765	False
Equity Maximum-Sharpe	fixed tilt / extended lexicon	0.488	0.731	False
Equity Maximum-Sharpe	adaptive tilt / extended lexicon	0.474	0.701	False
Equity Maximum-Sharpe	fixed tilt / plain lexicon	0.435	0.694	False

Tuning on 2021 and 2022 selects the adaptive tilt on the plain lexicon for both funds, and that choice also ranks first on 2023 - Maximum-Sharpe at 0.835 against 0.765 for the base fund. The ranking survived a period it was not chosen on, which is more than the tuned tilt in the course's own worked example managed.

Three caveats belong with that. The holdout margin over the base fund is a few hundredths of a Sharpe point, consistent with the bootstrap finding that the difference is not established. Every Minimum-Variance variant decays by roughly 0.58 of a Sharpe point into 2023, the base fund included, so the decay is the market regime and not a property of the tilt. And one split of two funds is evidence rather than proof. Bailey and Lopez de Prado (2014) show that the expected maximum Sharpe ratio across a set of trialled strategies rises with the number trialled even when none of them has any skill, which is why the number of variants searched is reported here alongside the one that won.

The sample-noise diagnosis in Section 2 is tested in Appendix E.

5.4 Fear and greed as a timing overlay

The sector tests ask which sector to overweight and say nothing about when to be in the market. The market-wide gauge answers that, and cannot be held, so it scales exposure instead: the fund moves toward cash as the gauge turns greedy, resets monthly, and never borrows.

It must be re-standardised first. Section 3 uses full-sample moments, which describe history correctly and are look-ahead if traded: a z-score in 2020 is measured against 2023.

Standardising on past observations only halves the raw signal - the spread between the most fearful and greediest deciles falls from 26.9 to 14.3 basis points a day. Look-ahead entered through a preprocessing step, not the backtest.

On the causal signal the overlay improves all four combined funds net of costs and cuts every drawdown. Risk Parity rises from 0.888 to 1.113 while its worst fall shrinks from 19.5% to 15.0%, holding 88% on average.

Shuffling the same exposures through time - same average and turnover, only the timing destroyed - gives a mean Sharpe of 0.600 over 1000 permutations, maximum 0.935; the real ordering reaches 1.113 and beats every one. The bootstrap difference is +0.225 [+0.011, +0.453], $p = 0.038$, and selecting on 2021-2022 gives 1.275 on 2023 against 1.206. It is the only result here clearing conventional significance.

About ten specifications were searched first, so the p-value fails a Bonferroni correction; the overlay makes 36 decisions; and the sample holds one large drawdown year, which flatters any de-risking rule. The claim is narrow: cross-sectional sentiment carries nothing tradable; the market-wide gauge carries something.

Table 10: The fear and greed exposure overlay on each combined fund, net of 10 bp of turnover cost. Out-of-sample, 2021-2023.

Fund	Sharpe base	Sharpe timed (net)	Drawdown base	Drawdown timed
Combined Equal-Weight	0.761	1.002	-0.279	-0.203
Combined Minimum-Variance	0.477	0.642	-0.154	-0.112
Combined Maximum-Sharpe	0.981	1.144	-0.258	-0.197
Combined Risk Parity	0.888	1.113	-0.195	-0.150

6. The app and the investor journey

Spotlight is a Streamlit application. A visitor compares the twelve funds on one table and chart, opens a fund's fact sheet for its growth of \$1, drawdown, current holdings and sector history, sets an allocation across funds with sliders and sees the blended track record, and reads the sentiment analytics including the fear and greed gauge and the sector index.

The application reads only precomputed CSV files committed under results/. It runs no backtest and loads no sentiment model, so cold starts stay fast on Streamlit Community Cloud's free tier. scripts/run_part_b.py rebuilds every artifact end to end.

The target user is a working professional or early-career investor with savings, no time, and no intention of reading a covariance matrix. The journey is built around the decision they actually make: not which stock to buy but how much risk to carry. They compare twelve funds, open a fact sheet for the four numbers a fund provider publishes, and set an allocation with sliders, where diversification stops being a word and becomes a visibly shallower drawdown. The analytics sit apart from the funds on purpose: they do not predict returns, and placing them beside an allocation control would imply a link the evidence does not support.

The business model is a management fee on assets, shown in the allocation tab.

Spotlight is deployed at <https://spotlight-z5640476.streamlit.app> from the public repository https://github.com/corvus-chen/z5640476_projectB. The app reads the committed artifacts and starts without recomputing anything, so it runs inside Community Cloud's free-tier resources.

7. Critical reflection and recommendations

What worked was a working order I set early and kept: the assistant explains an approach in full, I read it until I can correct it, and only then is code written. That is why the solver-scaling and volatility-drag problems are recorded as decisions rather than as bugs found later. The checks that came out of it earned their place - identity tests on the backtest exposed a real error, weights applied as fixed targets between rebalances, which is arithmetically a fund trading back to target daily and had inflated the best fund's Sharpe ratio from 0.98 to 1.03.

What did not work was the sentiment fusion, and my first explanation of it was wrong in an instructive way. I concluded that headline sentiment is contrarian, and every number I held supported that: eight negative sectors, a losing tilt, a profitable reversal. What I had not done was compare it against anything. A negative correlation fits a reversal, news already in the price, and noise equally well, and the one statistic I was reading cannot separate them. An explanation that fits all the evidence is not thereby correct.

The broader lesson is where the risk in this kind of project sits. None of the errors were errors of finance. A number was recalled instead of measured; a solver would have returned its starting point without complaining; a docstring described wrong behaviour accurately; a test suite only ever ran the defaults. Each had been checked, and the check is what gave the false confidence. The modelling was the easy part.

First, offer the adaptive sentiment tilt as an optional fund rather than as the default, and size the position to the strength of the evidence. The naive tilt should not ship at all: it makes both equity funds worse and charges turnover for it. The adaptive version is a different case - it improves both funds and ranks first on a holdout it was not chosen on, yet its bootstrap interval spans zero. Treating 'not established' as 'worthless' is as wrong as treating it as proven. I would hold it with part of my own money and not all of it, and the product should say exactly that: available, labelled with its evidence, never the default.

Second, do not pick a default fund at all. Make the user choose between Combined Maximum-Sharpe and Combined Risk Parity after the drawdown figure has been put in front of them. The temptation is to default the cautious user into risk parity, and I reject it: 0.888 against 0.981 is a real cost, and a platform making that trade quietly has decided how much risk someone can bear without asking. Show both - 24.0% a year with a 25.8% fall, or 14.0% with 19.5% - and make the choice deliberate. I would take the first for myself and would not make that decision for anyone else.

Third, hold a firm's evidence to a higher standard than my own. I would put part of my money behind the adaptive tilt on what this project shows; I would not put anyone else's there. The difference is not courage: a manager charging a fee claims to know something, and an interval spanning zero is not knowledge. Before this reaches a fund taking deposits, the same walk-forward design should run on a decade of data or a second market, with the number of specifications declared in advance rather than counted afterwards.

References

- Bailey, DH & Lopez de Prado, M 2014, 'The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality', *The Journal of Portfolio Management*, vol. 40, no. 5, pp. 94-107.
- Boudoukh, J, Richardson, M & Whitelaw, RF 2008, 'The myth of long-horizon predictability', *The Review of Financial Studies*, vol. 21, no. 4, pp. 1577-1605.
- Chopra, VK & Ziemba, WT 1993, 'The effect of errors in means, variances, and covariances on optimal portfolio choice', *The Journal of Portfolio Management*, vol. 19, no. 2, pp. 6-11.
- Da, Z, Engelberg, J & Gao, P 2015, 'The sum of all FEARS investor sentiment and asset prices', *The Review of Financial Studies*, vol. 28, no. 1, pp. 1-32.
- DeMiguel, V, Garlappi, L & Uppal, R 2009, 'Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy?', *The Review of Financial Studies*, vol. 22, no. 5, pp. 1915-1953.
- Fama, EF 1970, 'Efficient capital markets: a review of theory and empirical work', *The Journal of Finance*, vol. 25, no. 2, pp. 383-417.
- Harvey, CR, Liu, Y & Zhu, H 2016, '... and the cross-section of expected returns', *The Review of Financial Studies*, vol. 29, no. 1, pp. 5-68.
- Hutto, CJ & Gilbert, E 2014, 'VADER: a parsimonious rule-based model for sentiment analysis of social media text', *Proceedings of the Eighth International AAAI Conference on Weblogs and Social Media*, Ann Arbor, Michigan, pp. 216-225.

- Jagannathan, R & Ma, T 2003, 'Risk reduction in large portfolios: why imposing the wrong constraints helps', *The Journal of Finance*, vol. 58, no. 4, pp. 1651-1683.
- Ledoit, O & Wolf, M 2004, 'Honey, I shrunk the sample covariance matrix', *The Journal of Portfolio Management*, vol. 30, no. 4, pp. 110-119.
- Ledoit, O & Wolf, M 2008, 'Robust performance hypothesis testing with the Sharpe ratio', *Journal of Empirical Finance*, vol. 15, no. 5, pp. 850-859.
- Loughran, T & McDonald, B 2011, 'When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks', *The Journal of Finance*, vol. 66, no. 1, pp. 35-65.
- Maillard, S, Roncalli, T & Teiletche, J 2010, 'The properties of equally weighted risk contribution portfolios', *The Journal of Portfolio Management*, vol. 36, no. 4, pp. 60-70.
- Markowitz, H 1952, 'Portfolio selection', *The Journal of Finance*, vol. 7, no. 1, pp. 77-91.
- Michaud, RO 1989, 'The Markowitz optimization enigma: is 'optimized' optimal?', *Financial Analysts Journal*, vol. 45, no. 1, pp. 31-42.
- Okabe, M & Ito, K 2008, *Color universal design (CUD): how to make figures and presentations that are friendly to colorblind people*, viewed 11 August 2026, <<https://jfly.uni-koeln.de/color/>>.
- Politis, DN & Romano, JP 1994, 'The stationary bootstrap', *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303-1313.
- Sharpe, WF 1966, 'Mutual fund performance', *The Journal of Business*, vol. 39, no. 1, pp. 119-138.
- Sharpe, WF 1994, 'The Sharpe ratio', *The Journal of Portfolio Management*, vol. 21, no. 1, pp. 49-58.
- Tetlock, PC 2007, 'Giving content to investor sentiment: the role of media in the stock market', *The Journal of Finance*, vol. 62, no. 3, pp. 1139-1168.
- Tetlock, PC, Saar-Tsechansky, M & Macskassy, S 2008, 'More than words: quantifying language to measure firms' fundamentals', *The Journal of Finance*, vol. 63, no. 3, pp. 1437-1467.
- White, H 2000, 'A reality check for data snooping', *Econometrica*, vol. 68, no. 5, pp. 1097-1126.

Appendix

Appendix A - The four objectives

Each fund solves for a weight vector $\mathbf{w}$ over N assets from two estimated inputs: the vector of expected returns $\mathbf{m}$ and the covariance matrix $\mathbf{S}$, both computed on the estimation window and annualised. Every fund is long-only and fully invested, so $\mathbf{1}^T \mathbf{w} = 1$ and $\mathbf{w} \geq 0$.

(1) Equal weight: $w_i = \frac{1}{N}$ for every asset i $w_i = \frac{1}{N}$ for every asset i

(2) Minimum variance: $\min_{\mathbf{w}} \mathbf{w}^T \mathbf{\Sigma} \mathbf{w}$ subject to $\mathbf{1}^T \mathbf{w} = 1, \mathbf{w} \geq 0$ $\min_{\mathbf{w}} \mathbf{w}^T \mathbf{\Sigma} \mathbf{w}$ subject to $\mathbf{1}^T \mathbf{w} = 1, \mathbf{w} \geq 0$

(3) Maximum Sharpe: $\max_{\mathbf{w}} \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}}$ subject to $\mathbf{1}^T \mathbf{w} = 1, \mathbf{w} \geq 0$ $\max_{\mathbf{w}} \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}}$ subject to $\mathbf{1}^T \mathbf{w} = 1, \mathbf{w} \geq 0$

(4) Risk parity: $RC_i = \frac{w_i(\boldsymbol{\Sigma} \mathbf{w})_i}{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}} = \frac{1}{N}$ for all i $RC_i = \frac{w_i(\boldsymbol{\Sigma} \mathbf{w})_i}{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}} = \frac{1}{N}$ for all i

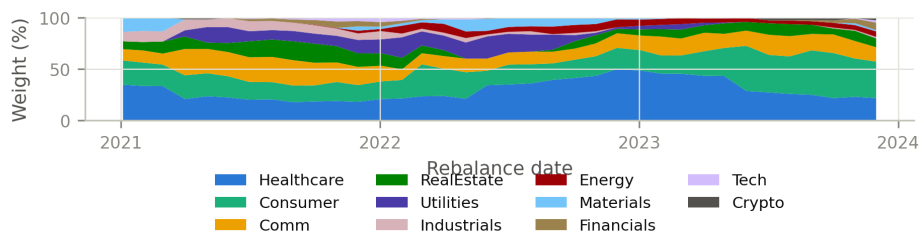
Here N is the number of assets in the family (50 equities, 10 cryptocurrencies, or 60 combined), w_i is the weight on asset i , $(\boldsymbol{\Sigma} \mathbf{w})_i$ is the i -th element of $\boldsymbol{\Sigma} \mathbf{w}$, RC_i is asset i 's risk contribution, and $\mathbf{1}$ is a vector of ones. Only equations (3) and (4) are affected by the choice of expected-return estimator; equation (2) uses S alone, which is why Section 5.4 can separate covariance noise from mean noise.

Appendix B - Supporting exhibits

Exhibits not needed in the main narrative.

Portfolio weights over time: Combined Minimum-Variance

Target weights set on the first trading day of each month, 04 Jan 2021 to 01 Dec 2023, aggregated to sector and stacked to 100%. Sectors are ordered by average weight, with crypto stacked last in dark grey. The weights move each month as the covariance matrix is re-estimated.



Source: course project data (equity and crypto daily adjusted closes, news headlines), 2020-2023. Out-of-sample backtest, author's calculations.

Figure 13: Sector weights over time for the Combined Minimum-Variance fund, the single-fund version of the across-methods exhibit.

Table 11: Full rebalance-frequency study: four schedules across the four combined funds, gross and net of 10 bp of one-way turnover.

method	frequency	n rebalances	sharpe	sharpe net costs	annual turnover
Equal-Weight	weekly	156	0.769	0.765	0.930
Equal-Weight	fortnightly	78	0.794	0.791	0.692
Equal-Weight	monthly	36	0.761	0.759	0.465
Equal-Weight	quarterly	12	0.725	0.724	0.270
Minimum-Variance	weekly	156	0.520	0.492	3.459
Minimum-Variance	fortnightly	78	0.499	0.479	2.497
Minimum-Variance	monthly	36	0.477	0.463	1.840
Minimum-Variance	quarterly	12	0.360	0.352	1.164
Maximum-Sharpe	weekly	156	0.971	0.939	7.669
Maximum-Sharpe	fortnightly	78	0.937	0.915	5.531
Maximum-Sharpe	monthly	36	0.981	0.966	3.926

Sharpe					
Maximum-Sharpe	quarterly	12	0.857	0.850	2.078
Risk Parity	weekly	156	0.902	0.896	0.905
Risk Parity	fortnightly	78	0.915	0.911	0.665
Risk Parity	monthly	36	0.888	0.885	0.456
Risk Parity	quarterly	12	0.828	0.827	0.283

Appendix C - The figure design system

Every exhibit is drawn through one design system rather than styled per figure. Categorical colours occupy fixed slots and are never cycled, so no two series in a chart can share a hue. The twelve-slot palette is validated pair by pair in CIE Lab space under normal, deuteranopic and protanopic vision by tools/check_palette.py; the worst separation is a deltaE of 16.9, identical to the original five-slot palette. The requirement that categorical series stay separable under simulated colour-blind vision, rather than only under normal vision, follows the colour-universal-design guidance of Okabe and Ito (2008).

The design system exists because the product is read by someone who will not check the code. A reader who cannot distinguish two bands in a weight chart cannot audit the fund's holdings, and a reader who is colour-blind should not be excluded from that. Fixing the colour slots and validating every pair under simulated colour-blind vision makes the exhibits verifiable rather than merely attractive.

Appendix D - Rolling against expanding estimation windows

Section 2 argues the estimation sample is too thin, which invites the question of why a longer one is not used. Each fund was therefore re-run with an expanding window that starts at the first observation instead of 252 days back, everything else unchanged. More history means less estimation noise but a longer memory of a regime that may have ended, and the two pull against each other. On equities, where the regime is comparatively stable, the extra data wins: minimum variance gains +0.110 and maximum Sharpe +0.090, while risk parity, which reads only the diagonal, is unmoved (-0.006). On the combined universe the memory costs more than the data is worth: Combined Maximum-Sharpe loses 0.583 of a Sharpe point, because an expanding window carries the 2020-21 crypto rally forward and keeps overweighting crypto into the 2022 drawdown. The rolling window is kept because the combined fund is the product. The main finding is unaffected either way: equal weighting at 0.819 still beats every optimised equity fund under both window types.

Table 12: Sharpe ratio under a rolling 252-day window against an expanding window, out of sample.

Family	Method	Rolling	Expanding	Change
Equity	Minimum-Variance	0.479	0.589	+0.110
Equity	Maximum-Sharpe	0.573	0.663	+0.090
Equity	Risk Parity	0.723	0.717	-0.006
Combined	Minimum-Variance	0.477	0.589	+0.112
Combined	Maximum-Sharpe	0.981	0.398	-0.583
Combined	Risk Parity	0.888	0.777	-0.111

The same test doubles as a check that this backtest is correctly implemented. The course's reference project estimates on an expanding window and publishes a Sharpe ratio for every fund, and those figures differ from the ones in Section 2 - most sharply for Combined Maximum-Sharpe, 0.40 there against 0.981 here. A gap that size would normally indicate an error. Re-running these funds on the reference window instead reproduces its numbers to a mean

absolute difference of 0.024, and to 0.002 on the fund where the headline figures diverge most. Equal weighting, which uses no estimation window at all, lands at 0.761 against its 0.76. The whole discrepancy is the window choice documented above, independently coded, and not a defect.

Table 13: These funds on an expanding window against the Sharpe ratios published by the course reference implementation.

Fund	Reference	This project, expanding	Difference
Combined Maximum-Sharpe	0.40	0.398	0.002
Combined Risk Parity	0.78	0.777	0.003
Combined Minimum-Variance	0.61	0.589	0.021
Equity Minimum-Variance	0.62	0.589	0.031
Equity Risk Parity	0.75	0.717	0.033
Equity Maximum-Sharpe	0.72	0.663	0.057

Appendix E - Does correcting the covariance fix the result?

The sample-noise half of that argument is testable, so it was tested rather than asserted. Ledoit-Wolf shrinkage (Ledoit and Wolf 2004) is the standard correction for a covariance matrix estimated from too few observations, and every fund was re-run with it. The three methods use the covariance differently, which makes the test discriminating: minimum variance is a pure function of the whole matrix, risk parity needs only its diagonal, and maximum Sharpe is dominated by the expected returns that shrinkage leaves untouched.

The results line up with that ordering. Minimum variance gains the most, +0.059 of a Sharpe point on the combined fund and +0.021 on the equity fund, and its holdings spread from 19 names to 27 as the extreme eigenvalues the optimiser had been leaning on are pulled back. Risk parity does not move at all (+0.000), which is what shrinking the off-diagonal terms of a matrix should do to a method that reads only the diagonal. Maximum Sharpe is mixed and small (+0.014 on equities, -0.022 combined), because its problem is the mean estimates and shrinkage does not address them. The correction helps exactly where the diagnosis said the noise was.

It does not, however, change the conclusion. Shrunk minimum variance still reaches only 0.500 against equal weighting's 0.819. Correcting the covariance recovers part of what sample noise costs and leaves the gap intact, which is the same result DeMiguel, Garlappi and Uppal (2009) report for the estimation-error corrections they test. The second force in the ordering - an objective that does not price return - is not something a better estimator can fix.

Table 14: Each fund re-run with a Ledoit-Wolf shrunk covariance matrix against the sample covariance. Out-of-sample, 2021-2023.

family	method	shrinkage	sharpe	ann vol	n assets held
Equity	Minimum-Variance	False	0.479	0.127	18
Equity	Minimum-Variance	True	0.500	0.126	19
Equity	Maximum-Sharpe	False	0.573	0.182	6
Equity	Maximum-Sharpe	True	0.587	0.182	6
Equity	Risk Parity	False	0.723	0.145	50
Equity	Risk Parity	True	0.723	0.145	50
Combined	Minimum-Variance	False	0.477	0.128	19

Combined	Minimum-Variance	True	0.536	0.126	27
Combined	Maximum-Sharpe	False	0.981	0.252	8
Combined	Maximum-Sharpe	True	0.960	0.254	8
Combined	Risk Parity	False	0.888	0.162	60
Combined	Risk Parity	True	0.888	0.162	60